

The Gram Point Hierarchy and the Lambert W Generator

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Abstract

The nontrivial zeros of the Riemann zeta function and their relationship to the Gram points are governed by a hierarchy of phase offset functions $\psi_k(t) = \vartheta(t) - k t$, where ϑ is the Riemann–Siegel theta function. The explicit closed-form expressions for the level- k Gram sequence $\{g_n^{(k)}\}$ is derived in terms of the Lambert W function and it is shown that all terms are generated by iterating a single base inversion operator Ψ composed with the Stirling residual tail of ϑ . The full Lagrange–Bürmann inversion produces an exact power series for $g_n^{(k)}$ in which every coefficient is expressible through $W_0\left((n + \frac{1}{8}) e^{-(2k+1)}\right)$, generated by a recurrence on polynomials p_j . The k -th level sequence $X_k(n) = n\pi - g_n^{(k-1)}$ is thereby expressed entirely in closed form through the same Lambert W generator at shifted level.

Table of contents

1	Definitions and Setup	1
2	The Explicit Gram Sequence	2
3	The Lambert W Generator for All Corrections	4

1 Definitions and Setup

Definition 1. [*Riemann–Siegel theta function*]

$$\vartheta(t) = \arg\left(\pi^{-it/2} \Gamma\left(\frac{1}{4} + \frac{it}{2}\right)\right) \quad (1)$$

normalized so that $\vartheta(t)$ is real and continuous for $t > 0$ with $\vartheta(0) = 0$.

Definition 2. [*Phase offset*] For each integer $k \geq 0$:

$$\psi_k(t) = \vartheta(t) - k t \quad (2)$$

Definition 3. [Stirling residual tail] The function $\varphi: (0, \infty) \rightarrow \mathbb{R}$ is defined by

$$\varphi(t) = \text{Im} \left[\log \Gamma \left(\frac{1}{4} + \frac{it}{2} \right) \right] - \frac{t}{2} \log \frac{t}{2} + \frac{t}{2} + \frac{\pi}{8} \quad (3)$$

so that

$$\vartheta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \varphi(t) \quad (4)$$

Consistency: $\vartheta(t) = \text{Im}[\log \Gamma(\frac{1}{4} + \frac{it}{2})] - \frac{t}{2} \log \pi$, and substituting (3) into (4) gives $\text{Im}[\log \Gamma(\frac{1}{4} + \frac{it}{2})] - \frac{t}{2} \log \pi$ since $\log \frac{t}{2\pi} - \log \frac{t}{2} = -\log \pi$.

The function φ is smooth on $(0, \infty)$ with asymptotic expansion

$$\varphi(t) \sim \frac{1}{48t} + \frac{7}{5760t^3} + \frac{31}{80640t^5} + \frac{127}{430080t^7} + \dots \quad (5)$$

arising from the Stirling series for $\log \Gamma$ at $z = \frac{1}{4} + \frac{it}{2}$ combined with the subleading terms of $\text{Im}[(z - \frac{1}{2}) \log z]$.

Definition 4. [Level- k Gram sequence] For each $k \geq 0$, the sequence $\{g_n^{(k)}\}_{n \geq 0}$ of positive reals satisfying

$$\psi_k(g_n^{(k)}) = n\pi, \quad n = 0, 1, 2, \dots \quad (6)$$

For $k=0$ this gives the ordinary Gram points $\vartheta(g_n^{(0)}) = n\pi$.

Definition 5. [Level sequence] For $k \geq 1$:

$$X_k(n) = \psi_k(g_n^{(k-1)}) = \vartheta(g_n^{(k-1)}) - k g_n^{(k-1)} \quad (7)$$

Definition 6. [Base inversion function] Let W_0 denote the principal branch of the Lambert W function. For $c > 0$ and integer $k \geq 0$:

$$\Psi(c, k) = \frac{2c}{W_0\left(\frac{c}{\pi} e^{-(2k+1)}\right)}. \quad (8)$$

2 The Explicit Gram Sequence

Theorem 7. [Explicit form of X_k] For every $k \geq 1$ and every $n \geq 0$:

$$X_k(n) = n\pi - g_n^{(k-1)}. \quad (9)$$

Proof. From (6) at level $k - 1$:

$$\vartheta(g_n^{(k-1)}) - (k-1) g_n^{(k-1)} = n \pi \quad (10)$$

, so

$$\vartheta(g_n^{(k-1)}) = n \pi + (k-1) g_n^{(k-1)} \quad (11)$$

Then (7) gives

$$\begin{aligned} X_k(n) &= n \pi + (k-1) g_n^{(k-1)} - k g_n^{(k-1)} \\ &= n \pi - g_n^{(k-1)} \end{aligned} \quad (12) \quad \square$$

Theorem 8. *[Exact Lambert W inversion of the Stirling bulk] Define*

$$f(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{(2k+1)}{2} t. \quad (13)$$

Then $\Psi(\cdot, k)$ inverts f exactly:

$$f(\Psi(c, k)) = c \quad \text{for all } c > 0, \quad (14)$$

and, with $c = \pi(n + \frac{1}{8})$,

$$g_n^{(k)} = \Psi(\pi(n + \frac{1}{8}), k) + O(n^{-1}) = \frac{2\pi(n + \frac{1}{8})}{W_0((n + \frac{1}{8}) e^{-(2k+1)})} + O(n^{-1}). \quad (15)$$

Proof. From (4), the Gram equation (6) is

$$f(t) + \varphi(t) = c \quad (16)$$

where

$$c = \pi(n + \frac{1}{8}) \quad (17)$$

The function f collects every term of $\vartheta(t) - kt$ that grows with t (the Stirling bulk), while $\varphi(t) = O(t^{-1})$ is the decaying Stirling residual. Crucially, the equation $f(t) = c$ is solved in closed form by $\Psi(c, k)$ with no approximation. To verify (14), substitute

$$t = \Psi(c, k) = \frac{2c}{w_k(c)} \quad (18)$$

into (13):

$$f(\Psi(c, k)) = f\left(\frac{2c}{w_k(c)}\right) = \frac{c}{w_k(c)} \log \frac{c}{\pi w_k(c)} - \frac{(2k+1)c}{w_k(c)} \quad (19)$$

The Lambert equation

$$w_k(c) e^{w_k(c)} = \frac{c}{\pi} e^{-(2k+1)} \quad (20)$$

gives

$$\log \frac{c}{\pi w_k(c)} = w_k(c) + (2k+1) \quad (21)$$

so

$$f\left(\frac{2c}{w_k(c)}\right) = \frac{c}{w_k(c)} (w_k(c) + (2k+1)) - \frac{(2k+1)c}{w_k(c)} = c \quad (22)$$

Thus $\Psi(c, k)$ is an exact root of $f(t) = c$.

The full Gram equation $f(t) + \varphi(t) = c$ can be rewritten as $f(t) = c - \varphi(t)$, a fixed-point problem $t = \Psi(c - \varphi(t), k)$ whose exact solution is given by the Lagrange–Bürmann series (Theorem 11). The $O(n^{-1})$ residual in (15) comes entirely from φ : since $\varphi(t) = O(t^{-1})$ and $\partial_c \Psi(c, k) = O(1/\log n)$, dropping φ incurs an error of order $O(n^{-1})$, not from any truncation of f or of W_0 . $\square$

Remark 9. The decomposition $\vartheta(t) - kt = f(t) + \varphi(t)$ separates the Gram equation into an exactly solvable part $f(t) = c$ and a perturbation φ . The Lambert W function absorbs the entirety of f —including all powers of $\log t$ —into a single closed-form expression $\Psi(c, k)$. No asymptotic expansion of W_0 is used; equation (14) holds identically. The Lagrange–Bürmann series in Theorem 11 then corrects for φ to all orders, yielding an exact expression for $g_n^{(k)}$ in which every coefficient is generated by Ψ and its derivatives.

3 The Lambert W Generator for All Corrections

Throughout this section, $w_k(c) = W_0\left(\frac{c}{\pi} e^{-(2k+1)}\right)$ and $a_k = \pi e^{2k+1}$, so that $\Psi(c, k) = 2c/w_k(c)$ and the Lambert equation reads $c = a_k w_k(c) e^{w_k(c)}$.

Theorem 10. *[Derivatives of Ψ] The j -th partial derivative of $\Psi(c, k)$ with respect to c is*

$$\partial_c^j \Psi(c, k) = \frac{2 e^{-(j-1) w_k(c)}}{a_k^{j-1}} \cdot \frac{p_j(w_k(c))}{(1 + w_k(c))^{2j-1}}, \quad (23)$$

where the polynomials p_j satisfy

$$p_1(w_k(c)) = 1 \quad (24)$$

$$p_{j+1}(w_k(c)) = p'_j(w_k(c)) (1 + w_k(c)) - ((j-1) w_k(c) + 3j-2) p_j(w_k(c)) \quad (25)$$

with first values

$$p_1(w_k(c)) = 1 \quad (26)$$

$$p_2(w_k(c)) = -1 \quad (27)$$

$$p_3(w_k(c)) = w_k(c) + 4 \quad (28)$$

$$p_4(w_k(c)) = -(2 w_k(c)^2 + 14 w_k(c) + 27) \quad (29)$$

Proof. From the Lambert equation $w_k(c) e^{w_k(c)} = c/a_k$, differentiating both sides with respect to c :

$$(1 + w_k(c)) e^{w_k(c)} \partial_c w_k(c) = \frac{1}{a_k} \quad (30)$$

giving

$$\partial_c w_k(c) = \frac{1}{a_k e^{w_k(c)} (1 + w_k(c))} \quad (31)$$

Base case ($j = 1$). From $\Psi(c, k) = 2c/w_k(c)$:

$$\partial_c \Psi(c, k) = \frac{2}{w_k(c)} - \frac{2c}{w_k(c)^2} \partial_c w_k(c) \quad (32)$$

Substituting $c = a_k w_k(c) e^{w_k(c)}$ and (31):

$$\begin{aligned} \partial_c \Psi(c, k) &= \frac{2}{w_k(c)} - \frac{2 a_k w_k(c) e^{w_k(c)}}{w_k(c)^2 a_k e^{w_k(c)} (1 + w_k(c))} \\ &= \frac{2}{w_k(c)} - \frac{2}{w_k(c) (1 + w_k(c))} = \frac{2}{1 + w_k(c)} \end{aligned} \quad (33)$$

confirming (23) at $j = 1$ with $p_1 = 1$.

Induction step. Assume (23) holds at index j . The product rule applied to

$$\partial_c^j \Psi(c, k) = 2 a_k^{1-j} e^{-(j-1)w_k(c)} p_j(w_k(c)) (1 + w_k(c))^{-(2j-1)} \quad (34)$$

and the factor $\partial_c w_k(c)$ from (31) yields:

$$\begin{aligned} \partial_c^{j+1} \Psi(c, k) &= \left(p_j'(w_k(c)) - (j-1) p_j(w_k(c)) - \frac{(2j-1) p_j(w_k(c))}{1 + w_k(c)} \right) \frac{2 e^{-(j-1)w_k(c)}}{a_k^{j-1} (1 + w_k(c))^{2j-1}} \\ &= \frac{2 e^{-j w_k(c)} p_j'(w_k(c)) (1 + w_k(c)) - ((j-1) (1 + w_k(c)) + (2j-1)) p_j(w_k(c))}{a_k^j (1 + w_k(c))^{2j+1}} \end{aligned} \quad (35)$$

The induction is closed by defining

$$(j-1) (1 + w_k(c)) + (2j-1) = (j-1) w_k(c) + 3j - 2 \quad (36)$$

since

$$p_{j+1}(w_k(c)) = p_j'(w_k(c)) (1 + w_k(c)) - ((j-1) w_k(c) + 3j - 2) p_j(w_k(c)) \quad (37)$$

The first 3 listed values follow:

$$p_2(w_k(c)) = 0 \cdot (1 + w_k(c)) - 1 \cdot p_1(w_k(c)) = -1 \quad (38)$$

$$p_3(w_k(c)) = 0 \cdot (1 + w_k(c)) - (w_k(c) + 4) \cdot (-1) = w_k(c) + 4 \quad (39)$$

$$p_4(w_k(c)) = (1 + w_k(c)) - (2 w_k(c) + 7) (w_k(c) + 4) = -(2 w_k(c)^2 + 14 w_k(c) + 27) \quad (40)$$

□

Theorem 11. *[Exact Gram sequence via Lagrange–Bürmann inversion] Define the correction function $\eta_k(c) = \varphi(\Psi(c, k))$. The exact solution to (6) is*

$$g_n^{(k)} = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} \frac{d^{j-1}}{d c^{j-1}} [\partial_c \Psi(c, k) \cdot \eta_k(c)^j]_{c=\pi(n+\frac{1}{8})} \quad (41)$$

where $\frac{d^{-1}}{d c^{-1}}$ at $j=0$ denotes the identity operator, all derivatives of Ψ are given by (23), and $\partial_c \Psi(c, k) = 2/(1 + w_k(c))$.

Proof. The Gram equation $f(t) = c - \varphi(t)$ with (14) gives the fixed-point equation

$$t = \Psi(c - \varphi(t), k). \quad (42)$$

By the Lagrange–Bürmann theorem (Whittaker–Watson, §7.32): if $t = \Psi(c - \varphi(t), k)$ and $t_0 = \Psi(c, k)$, then for any analytic F ,

$$F(t) = F(t_0) + \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \frac{d^{n-1}}{d c^{n-1}} [F'(t_0) \partial_c \Psi(c, k) \eta_k(c)^n] \quad (43)$$

Taking F to be the identity ($F(t) = t$, $F' = 1$) gives (41).

Convergence: the nearest singularity of $\Psi(\cdot, k)$ in the complex c -plane is the branch point of W_0 at $c = -a_k/e$, giving convergence radius $|\pi(n + \frac{1}{8}) + a_k/e| > \pi(n + \frac{1}{8})$ for $n \geq 0$. Since $|\eta_k(c)| = |\varphi(\Psi(c, k))| = O(c^{-1})$ while this radius grows linearly in c , the series converges for all $n \geq 0$. $\square$

Corollary 12. *[First correction term and second-order structure]*

$$g_n^{(k)} = \Psi(\pi(n + \frac{1}{8}), k) - \frac{2 \eta_k(\pi(n + \frac{1}{8}))}{1 + w_k(\pi(n + \frac{1}{8}))} + \frac{1}{2} \frac{d}{d c} \left[\frac{2 \eta_k(c)^2}{1 + w_k(c)} \right]_{c=\pi(n+\frac{1}{8})} + O(\eta_k(\pi(n + \frac{1}{8}))^3) \quad (44)$$

The $j=2$ derivative expands via the product rule to

$$\frac{1}{2} \frac{d}{d c} \left[\frac{2 \eta_k(c)^2}{1 + w_k(c)} \right] = -\frac{e^{-w_k(c)} \eta_k(c)^2}{a_k (1 + w_k(c))^3} + \frac{4 \eta_k(c) \varphi'(\Psi(c, k))}{(1 + w_k(c))^2} \quad (45)$$

where $\varphi'(t) = \frac{d}{dt} \varphi(t)$. Both summands in (45) are $O(\eta_k(c) / \Psi(c, k))$ and contribute at the same asymptotic order.

Proof. The $j=0$ term of (41) gives $\Psi(\pi(n + \frac{1}{8}), k)$. The $j=1$ term gives $-\partial_c \Psi(c, k) \cdot \eta_k(c) = -2 \eta_k(c) / (1 + w_k(c))$. The $j=2$ term is $\frac{1}{2} \frac{d}{d c} [\partial_c \Psi(c, k) \cdot \eta_k(c)^2]$. By the product rule:

$$\frac{1}{2} \frac{d}{d c} \left[\frac{2 \eta_k(c)^2}{1 + w_k(c)} \right] = \frac{1}{2} \partial_c^2 \Psi(c, k) \eta_k(c)^2 + \partial_c \Psi(c, k) \eta_k(c) \eta'_k(c). \quad (46)$$

From Theorem 10 at $j=2$: $\partial_c^2 \Psi(c, k) = -2 e^{-w_k(c)} / (a_k (1 + w_k(c))^3)$. The chain rule gives $\eta'_k(c) = \varphi'(\Psi(c, k)) \cdot \partial_c \Psi(c, k) = 2 \varphi'(\Psi(c, k)) / (1 + w_k(c))$. Substituting:

$$\frac{1}{2} \left(\frac{-2 e^{-w_k(c)}}{a_k (1 + w_k(c))^3} \right) \eta_k(c)^2 + \frac{2}{1 + w_k(c)} \cdot \eta_k(c) \cdot \frac{2 \varphi'(\Psi(c, k))}{1 + w_k(c)} \quad (47)$$

yields (45). □

Corollary 13. *[Explicit k -th level sequence] The k -th level sequence*

$$X_k(n) = n \pi - g_n^{(k-1)} \quad (48)$$

is

$$X_k(n) = n \pi - \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} \frac{d^{j-1}}{d c^{j-1}} \left(\frac{2 \eta_{k-1}(\pi(n + \frac{1}{8}))^j}{1 + w_{k-1}(\pi(n + \frac{1}{8}))} \right) \quad (49)$$

with

$$w_{k-1}(c) = W_0\left(\frac{c}{\pi} e^{-(2k-1)}\right) \quad (50)$$

and

$$\eta_{k-1}(c) = \varphi(\Psi(c, k-1)) \quad (51)$$

For $k=1$, through first-order correction:

$$X_1(n) = n \pi - \frac{2 \pi (n + \frac{1}{8})}{W_0((n + \frac{1}{8})/e)} + \frac{2 \varphi(\Psi(\pi(n + \frac{1}{8})))}{1 + W_0((n + \frac{1}{8})/e)} + O(\varphi^2) \quad (52)$$

Proof. Substitute (41) with k replaced by $k-1$ into (9). □

Remark 14. The polynomials p_j in (24) depend only on $w_k(c)$ and not on k . The factor $e^{-(j-1)w_k(c)}$ in (23) introduces $w_k(c)$ -dependence beyond the polynomial, so the full j -th derivative is not purely a rational function of $w_k(c)$ and $1 + w_k(c)$. The dependence on the level k enters through the argument of W_0 : replacing k by $k+1$ multiplies the argument by e^{-2} . Every level of the hierarchy is generated by the same algebraic structure—(24) and Ψ —evaluated at a geometrically rescaled argument of W_0 .